# BEST KNOWN SQUARE PACKINGS

**The square packing problem asks for the side $s(n)$ of the smallest square that can hold $n$ unit squares, where the squares are free to rotate but cannot overlap**

- O proved optimal (77 of 324)
- = exact value known (307 of 324)
- ≈ only known numerically (17 of 324)
- R rigid (22 of 324)

90° 45° colors indicate distinct tilt angles

4 3 2 1 0 shade indicates number of full-side contacts

deg is the algebraic degree of that side length

recent result (since August, 2026)

Best packings due to Göbel, Trump, Wainwright, Hämäläinen, Stenlund, Friedman, Bidwell, Cantrell, Morandi, Ellsworth, Hajba, Schadt, Chaoweeraprasit, Couzo, Daniel, Fang, Ostashev, PingYou Ltd, Rehwaldt, Gupta, Barnett, ry-xu, Chang, DeVincentis, hmbelvedere, Mishapolk

Lower bounds due to Göbel, Stromquist, Kearney, Shiu, Bentz, Daniel, Vlasenko, Guzhou0806, Hosono, Ahmed, Karakuş, Ryu

Optimality proofs due to Göbel, Stromquist, Friedman, El Moumni, Kearney, Shiu, Bentz, Vlasenko, Levy, Daniel, Hosono, Ahmed, Karakuş; formalization contributions by Gurevitch, Guzhou0806; verification infrastructure and execution by EvolvingPrograms, Julian-JJ, Lewis

Diagram by Joshua Levy

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The Squares Project

github.com/jlevy/squares